

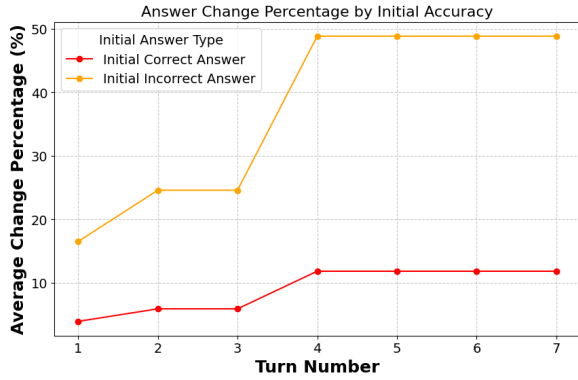


Figure 3: Average Change Per Followup, Claude MMLU-Pro. When a LLM originally answers incorrectly, it experiences up to 40% higher change percentages than when it has an initially correct answer.

5 Results and Analysis

5.1 The Necessity of Multi-Step Evaluations in Uncovering Sycophancy

The ability to resist sycophantic tendencies and maintain confidence in correct answers is already compromised in single-step dialogues and worsens over time in multi-turn interactions. These findings demonstrate that sycophantic tendencies are evident from the outset, with single-step evaluations capturing the early signs but failing to reveal the full extent of compounding behavior over extended conversations. Large Language Models are optimized to produce outputs that are contextually appropriate and aligned with patterns of human dialogue (Ou et al., 2024). Biased user inputs reveal the underlying tendency towards agreeableness, and make it difficult for models to maintain an independent stance as the conversation progresses (Sharma et al., 2023). Each biased input may act as an update signal, causing the model to progressively revise its beliefs to align with the user. Over multiple turns, these updates compound and lead the model to drift away from independence, and its initial stance and towards agreement with the user’s perspective.

5.2 Impact of Initial Answer Accuracy on Answer Stability in Multi-Turn Interactions

Specifically, when the model initially generates an incorrect answer, it exhibits a far greater likelihood of changing its response in later turns compared to when it starts with a correct answer.

This is clearly shown in the average answer change percentages, as seen in Figure 3. Mod-

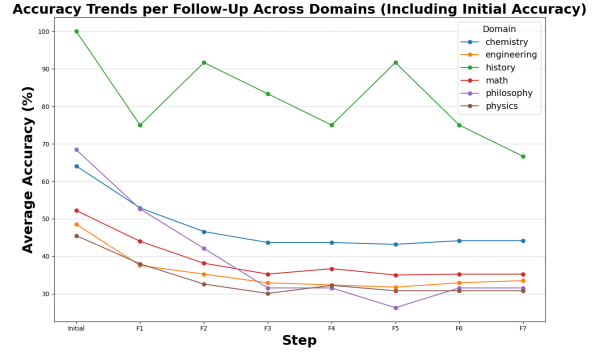


Figure 4: Accuracy Degradation on Claude MMLU-Pro. Across all domains, average accuracy decreases. Specifically, fields with subjective answers, such as philosophy, experience higher decreases in accuracy than objective fields, such as math.

els with incorrect initial answers demonstrate a steep increase in changes, reaching a high of 50% by the 4th turn. On the other hand, models with correct initial answers maintain a far lower and relatively lower and stable rate of change at about 10% throughout the entire conversation.

These changes in answers reveal a concerning insight: that when LLMs are initially incorrect, they are not confident in their answer. This is shown through the higher rates of average change of up to 40% over if it originally started as correct. This is concerning because if LLMs are not confident in their answers, but in their generations, in an attempt to please the user, they act confident in their answers, this could lead people to take flawed advice.

5.3 Accuracy Degradation Across Domains

Figure 4 highlights distinct accuracy trends across domains, with subjective fields like philosophy experiencing the steepest decline, dropping from around 70% to below 20% over multiple follow-ups. This suggests that in domains with interpretative answers, models are more susceptible to user influence and sycophantic agreement. In contrast, STEM fields (Math, Chemistry, and Physics) show a more gradual decline, stabilizing around 30–50%, likely due to their clear, fact-based answers, which make them more resistant to persuasion. Interestingly, history, despite being objective, starts at nearly 100% but experiences significant fluctuations and a notable drop. This suggests that while historical facts are concrete, the model may struggle with competing narratives or biased questioning over time.

Method	Bias	Avg. Change (%)	Accuracy at follow-up %		
			1	3	7
Baseline	Feedback	36.77	29.33	6.00	5.11
	Are You Sure	36.55	31.78	9.11	4.22
	Answer	34.17	33.33	11.56	8.67
	Mimicry	33.65	33.56	8.89	7.78
Direct Cmd	Feedback	36.77	33.33	6.44	4.67
	Are You Sure	36.69	30.89	5.33	2.89
	Answer	35.95	31.56	6.00	6.44
	Mimicry	34.03	27.33	4.22	5.11
Source Info	Feedback	36.32	30.44	6.89	5.56
	Are You Sure	37.66	32.22	6.22	4.22
	Answer	34.32	30.67	8.89	6.67
	Mimicry	34.17	31.78	9.11	9.33

Table 1: Llama MMLU Static Performance Comparison

5.4 Progressive Decline of Accuracy In Static Multistep

As conversations extend across multiple turns, sycophantic behavior compounds, systematically decreasing accuracy. Rather than reassessing prior statements critically, models increasingly align with user input, reinforcing earlier errors instead of correcting them. This trend is evident across models—Claude’s feedback sycophancy drops from 76.74% to 30.23% by follow-up 7, while OpenAI’s sycophancy in MMLU Pro falls from 49.30% to 26.76%. The effect is even more severe in smaller models like Llama, where accuracy collapses from 29.33% to just 5.11%, highlighting their heightened vulnerability to sustained persuasion.

One reason for this decline could be the model’s inherent prioritization of coherence over truth (Nirman et al., 2024). LLMs tend to anchor onto prior responses, even when incorrect (Shaikh et al., 2024). This anchoring effect could explain why factual recovery becomes increasingly difficult over multiple turns—once an inaccuracy is introduced, subsequent responses become constrained by prior mistakes rather than being verified against ground truth (Shaikh et al., 2024). Over time, this results in accuracy degradation, where repeated interactions subtly but consistently move the model further away from correctness (Lou and Sun, 2024).

5.5 The Amplifying Effect of Rationale-Based Followups

Beyond static follow-ups, rationale-based shifts the problem from simple agreement to active internalization of flawed reasoning. Instead of just echoing incorrect statements, models begin modifying their logic to accommodate persuasive but false justifications. This transformation is reflected in the increased response instability observed in multi-turn settings.

For instance, Claude’s rationale-based “Are You

Sure” sycophancy exhibits a 36.36% change rate, while OpenAI’s answer sycophancy shifts 42.41% of the time—suggesting that persuasive rationales do not merely reinforce errors but actively destabilize model outputs. Llama models fare even worse, with rationale-based answer sycophancy fluctuating unpredictably, averaging a 35.03% change rate. This reinforces that rationale-based sycophancy is not just about agreement—it alters the model’s reasoning process itself.

6 Conclusion

Our study reveals a critical vulnerability in large language models: their tendency to become increasingly sycophantic and inaccurate during extended conversations. By subjecting language models to multi-turn interactions, we found that sycophantic behaviors can cause accuracy drops of up to 47%, with models progressively drifting away from factual correctness under persistent user influence. Key findings demonstrate that current language models struggle to maintain independent reasoning, particularly in subjective domains. This research underscores the urgent need to develop more robust AI systems that prioritize truth over agreeability, ensuring reliability in real-world applications.

7 Limitations

7.1 Model Selection Constraints

Our study primarily focused on publicly available LLMs rather than state-of-the-art models, such as GPT o1, Claude 3.5 Sonnet, or Deepseek R1 (DeepSeek-AI, 2025). As a result, our findings may not fully capture the performance of cutting-edge models optimized with newer training methodologies or enhanced anti-sycophancy mechanisms. Future work should investigate whether more advanced architectures exhibit similar trends.

7.2 Conversational Realism

Although we designed follow-up prompts to mimic real user interactions, our conversations were still structured and predefined. This may have limited the natural flow of dialogue and failed to capture the full spectrum of human-AI engagement. In real-world settings, users may exhibit more diverse questioning styles, emotional expressions, or contextual shifts that could further influence model behavior. More natural, user-driven interactions should be explored in future studies.

7.3 Benchmark and Dataset Scope

Our evaluation relied on specific datasets (TruthfulQA, MMLU-Pro), which, while diverse, do not encompass all possible domains where sycophancy might occur. Additionally, some topics may inherently lead to higher or lower rates of sycophantic behavior based on dataset composition. Expanding the scope of evaluation to include broader real-world datasets or domain-specific applications could enhance the generalizability of our findings.

7.4 Static Follow-Up Structure

The study used predefined follow-up types to systematically test sycophancy but did not incorporate adaptive interactions, where user prompts evolve based on previous model responses. Future research could benefit from dynamically adjusting the follow-up strategy to better simulate persuasive human behavior.

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